



GOA UNIVERSITY

B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)**Paper Title: Data Analytics (Professional Elective)****Paper Code: CE 723****[Duration: 3 Hours]****[Total Marks:100]**

- Instructions: i) Attempt any two questions from PART A and any two questions from PART-B and any one question from PART-C.
 ii) Assume suitable data if necessary.

PART A

- Q.1** a) The weights of 8 individuals were recorded before and after participating in a fitness program. Is there any significant difference between their weights before and after the program? Analyse. Use $\alpha=0.05$ and t value=2.36 7

Before	180	200	230	240	170	190	200	165
After	140	145	150	155	120	130	140	130

- b) Find a Linear Regression equation for the following 2 sets of data

X	2	4	6	8
Y	3	7	5	10

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- c) Practical test marks of students are as follows: 10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9 7
 Mean population score is 12 and standard deviation is 1.80. Is the mean value for student significantly differed from the mean population value? Let alpha value is 0.05, so corresponding t value is 2.262.

- Q.2** a) In a public health campaign aimed at reducing the incidence of a particular disease, a new medication was administered to 812 people out of a total of 3248. The number of disease cases is shown below. $X^2_{0.05}=3.84$. Determine whether there is a relationship between receiving the medication and the occurrence of the disease. **5**

Medication	Disease(A)	No Disease (a)	Total
Received(B)	140	30	170
Not Received (b)	60	20	80
Total	200	50	250

- b) Calculate χ^2 value using the table below. which represents the observed and expected frequencies of individuals based on their preferred mode of transportation and gender. Critical $\chi^2 = 7.851$ **7**

Mode of Transportation	Observed Frequency		Expected Frequency	
	Female	Male	Female	Male
Public Transport	7	8	6.08	6.92
Personal Vehicle	14	17	13.55	15.45
Bicycle	15	14	14.49	16.51
Walking	7	10	8.88	10.21

- c) Find the value of t-test value for the following set of values: **4**

7, 2, 9, 8 and 1, 2, 3, 4

- d) Write short notes on: **4**
- Logistic Regression
 - ANOVA (Analysis of Variance)
 - Hypothesis Testing
 - Probability of Error

- Q.3** a) A sample of 16 employees is taken to investigate the average number of sick days taken in a year. A claim is made that the average number of sick days is more than or equal to 5 days. Based on the sample, the mean number of sick days was found to be 9 days, with a sample standard deviation of 3 days. At a significance level of 0.05, the t-value for a right- tailed test is 1.753. Calculate the t-value and check whether the null hypothesis holds. **6**

- b) The subject marks of boys and girls are as follows:

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Boys: 12, 14, 10, 8, 16, 5, 3, 9, 11

Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 15

Is there any significant difference between the marks of girls and boys? Assume $\alpha=0.05$, t-value (critical) = 2.11.

- c) To assess the significance of possible variation in a certain test between the convent schools of a city, a common test was given to a number of students taken at random from the fifth class of 3 schools concerned, the results are given below.

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$$F_{0.05} = 3.89$$

School A	School B	School C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Make analysis of variance for the given data.

PART B

- Q.4** a) Compare and contrast structured data and unstructured data.

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- b) Describe four different methods for creating a matrix in the R programming language, including examples for each.

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- c) Explain different types of NoSql database.

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- d) Explain how a decision tree model is built using R programming language. Also show commands to plot a tree.

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- Q.5** a) Create a data frame named "Employee" in R programming language with three attributes (Employee ID, Name, Department). Also display:

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- All the elements in the first row
- All the elements of the first column
- Element at the 2nd row, 3rd column

- b) Explain different ways of reading data in R.

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- c) Explain all the control statements available in R Programming Language with the help of an example.

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- Q.6** a) Compare MySql with NoSql database. 6
- b) Explain each phases of data analytic life cycle. 7
- c) Explain the different data types supported in R with the help of examples. 7

PART-C

- Q.7** a) Explain different types of database languages. 7
- b) Explain at least three data visualization function in R programming language with the help of an example. 6
- c) Explain database system structure with a neat diagram. 7
- Q.8** a) The following data represents the productivity scores of employees from three different departments. Test the hypothesis that the mean productivity scores across the three departments are equal. $F_{0.05} = 3.68$ 8

Department A	Department B	Department C
8	8	17
10	6	10
7	11	12
14	8	12
11	8	15
16	13	12

- b) Write short notes on Significance level, Directed hypothesis, Probability of error, Hypothesis test. 5
- c) The Internal Test 1 marks of B.E Computer Boys and Girls are as follows: Boys: 12, 13, 14, 10, 18, 16, 5, 3, 9, 4, 11; Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 4, 15. Is there any significant difference between marks of girls and boys? Analyse by considering $\alpha = 0.05$ and t-value = 2.11 5
- d) Discuss in detail Chi^2 test with detail example. 2